A Novel DEEP-SN Hybrid Approach for Segmenting and Classifying Solar Panel

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***Abstract*— The detection of solar panels is vital for optimizing the efficiency of solar energy systems. This study presents a novel approach for solar panel detection using a hybrid deep learning model, DEEP-SN. The model combines the segmentation power of DeepLabV3 with the classification capabilities of the Swin Transformer. DeepLabV3 leverages a ResNet50 backbone and atrous spatial pyramid pooling (ASPP) to capture multi-scale information for precise image segmentation. With its ability to process long-range dependencies and spatial hierarchies, the Swin Transformer enhances the model's classification accuracy. We employ a class-weighted loss function to han- dle a class imbalance in the dataset, and the OneCycleLR scheduler dynamically adjusts the learning rate during training. The model achieves a validation accuracy of 90%, a test accuracy of 90%, and an F1 score of 0.90, reflecting its strong performance in detecting solar panels. This approach offers a promising solution for automating solar panel detection, contributing to more effective monitoring and management of solar energy systems.**

Keywords—Solar Panels, DEEP-SN, Detection, DeeplabV3, Classification.

I. INTRODUCTION

Solar photovoltaic (PV) systems must be rapidly adopted to mitigate climate change and support sustainable energy. Large-scale deployment and maintenance depend on accurate detection, segmentation, and classification to ensure optimal performance. Traditional methods such as manual inspections are inefficient and labor-intensive, driving the need for advanced machine learning solutions.

Convolutional Neural Networks (CNNs) are pivotal for automating PV identification. A U-Net-based CNN achieved 94% accuracy and a 0.64 IoU for rooftop PV mapping in Switzerland [1]. Another U-Net model using augmentation and weighted loss attained a 91.42% dice score and 90.42% IoU [2]. TEMCA-Net, integrating texture enhancement and ASPP, achieved a 92.21% mean IoU while reducing false positives in complex ur- ban scenes [3]. LIRNet, a lightweight inception-residual network, enables real-time defect detection with 89% accuracy [4].

Machine learning classifiers like Random Forest and XGBoost, using spectral and terrain data, have demon-strated 99.65% accuracy in large-scale PV mapping across China [5]. In Manchester, hybrid frameworks like Mask R-CNN outperformed SegNet with 95.66% accuracy using high-resolution satellite imagery [6]. The PKGPVN model employs Photovoltaic Index (PVI) and hybrid attention to further refine segmentation and boundary delineation [7].

Transformer-based models such as PV-YOLO surpass traditional YOLOs, achieving 92.56% mean average precision in detecting small PV defects [8]. Multi- resolution models like DeepLabV3 have shown strong generalizability, with 91.04% IoU and 95.27% F1-score [9].

Thermal imaging is gaining traction due to its robustness in harsh conditions. A Mask R-CNN trained on UAV-captured thermal data reached a recall of 98.1% [10]. HyperionSolarNet, combining EfficientNet-B7 with segmentation, achieved 96% classification ac- curacy and 82% IoU [11]. Transfer learning-based U-Nets have achieved 99% F1-scores and 98% damage classification accuracy in maintenance tasks [12], while outperforming other models in soiling detection with an 86.3% Jaccard Index [13].

Lightweight models such as Deep-SN are ideal for large-scale PV deployments due to their efficiency and segmentation performance [14], [15]. Additionally, an ANN-based model using three years of meteorological data and MATLAB/Simulink reached 99.4% prediction accuracy for solar energy forecasting [16].

Building on this foundation, we propose the DEEP-SN hybrid model to address dataset variability and class im-balance. Its novel architecture and robust preprocessing yield superior performance across all metrics, supporting scalable solar PV detection and classification.

- A dataset of 4,616 images was used, with pre-processing techniques including resizing, augmentation (flipping, rotation, color jittering, cropping), normalization, and class balancing. These ensured data consistency, efficient training, and improved generalization.
- DEEP-SN was benchmarked against multiple CNN-based object detection models. It consistently outperformed all alternatives, demonstrating higher accuracy and reliability in solar panel detection.

The paper is organized as follows: Section II presents the proposed methodology and system architecture. Section III discusses experimental results and model evalation. Section IV concludes with a summary of findings and the significance of DEEP-SN in PV detection and classification.

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II. Proposed Method

The goal of solar panel detection is achieved through a structured approach comprising three stages, as illustrated in Fig. 1. The first stage involves the collection of high-resolution solar panel images, enhanced with advanced preprocessing techniques to ensure data quality. The second stage focuses on segmenting and classifying solar panels using the DEEP-SN, a hybrid model that combines DeepLabV3 for precise segmentation and the Swin Transformer for classification. Finally, the third stage evaluates the model's performance using metrics like accuracy and computational efficiency, assessing its impact on solar panel efficiency.

This framework refines segmented images to detect even the smallest irregularities, such as cracks, dust accumulation, and structural wear, which can affect energy efficiency. By leveraging the Swin Transformer, the model captures intricate patterns at different scales, ensuring a deep understanding of panel conditions. Segmented outputs highlight areas of concern, allowing for precise detection of cracks, dust accumulation, or structural wear that could impact energy efficiency. These refined results undergo rigorous evaluation using performance indicators such as ROC curves, line plots, and confusion matrices, ensuring the model's reliability in real-world applications. Additionally, computational efficiency is carefully assessed to balance accuracy and processing speed, making the system well-suited for automated solar panel monitoring. This structured methodology enhances defect detection and enables proactive maintenance, ultimately improving solar panel performance and longevity.

A. *Data Acquisition and Preprocessing*

We selected a publicly available dataset to validate our research methodology. The dataset contains 4,616 images, with 3230 allocated for training, 692 for validation, and 694 for testing. The images were divided into training, validation, and testing sets in a 75%-15%-15% ratio.

In the preprocessing stage of the solar panel detection framework, several strategies were implemented to prepare the data for optimal model performance. The images, originally high-resolution, were resized to a standard 224 × 224 pixels to ensure consistency across the dataset. To further enhance model generalization, we applied a range of augmentation techniques such as random horizontal and vertical flips, rotations, and color jittering, which helped the model learn to handle variations in image orientation and lighting. Additionally, random cropping and resizing were used to simulate different perspectives, improving the model's ability to adapt to changes in object size and location. To ensure the images were appropriately scaled, pixel values were normalized using ImageNet's mean and standard deviation. Addressing class imbalance was crucial, so stratified sampling and class-weighted loss functions were incorporated to balance the learning process. Prefetching during training also helped optimize the data loading process, ensuring efficient batch processing. These preprocessing steps collectively improved the dataset's diversity and robustness, setting a strong foundation for the subsequent stages of segmentation and classification.

Lastly, the dataset was split into training, validation, and test sets. The training set was used for model fitting, the validation set for hyperparameter tuning, and the test set for final performance evaluation, ensuring reliable and accurate classification of solar panels.

B. *Proposed DEEP-SN Model*

The proposed model combines DeepLabV3 for segmentation and Swin Transformer for classification, specifically designed for solar panel detection as shown in the part B. Architecture of Fig. 1. DeepLabV3 introduced [17], is a powerful model for semantic image segmentation, particularly effective for identifying solar panels. It uses a ResNet50 backbone in the encoder for feature extraction, progressively reducing spatial resolution while preserving essential low-level and high-level features crucial for detecting solar panels. The encoder features layers of DeepLabV3, including the initial convolution block at 128x128 resolution, followed by successive inception blocks (1, 2, and 3) at 64x64, 32x32, and 16x16 resolutions. The Atrous Blocks (1 and 2) reduce the resolution to 8x8 and 4x4, allowing the model to capture multi-scale context and effectively distinguish solar panels based on size and location. The Atrous Spatial Pyramid Pooling (ASPP) module refines the model's ability to capture different scales, resulting in a 4x4 output. In the decoder, the feature maps are up-sampled through five stages to reconstruct the segmentation map, with resolutions progressing from 4x4 to 128x128, ensuring precise and detailed solar panel detection. This segmentation map, which is also shown in Figure 3, is then passed to the Swin Transformer, a powerful hierarchical vision transformer introduced [18]. The Swin Transformer excels in capturing long-range dependencies and understanding spatial hierarchies. Its high performance in vision tasks makes it ideal for classifying the solar panels detected by DeepLabV3, ensuring robust and accurate classification. Combining DeepLabV3's segmentation capabilities with Swin's advanced classification power, the model effectively detects and classifies solar panel images.

C. *Optimization and Evaluation Factors*

For the optimization and evaluation of this hybrid model. The AdamW optimizer adjusts the learning rate dynamically based on the gradient of the loss, facilitating quicker and more stable convergence of the model.

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} + \lambda \cdot \theta_t \qquad (1)$$

Additionally, the OneCycleLR scheduler dynamically modifies the learning rate during training, starting with a low value, increasing it to a peak, and then reducing it gradually. This ensures the optimizer explores the parameter space effectively and avoids overfitting by settling into a stable configuration.

$$\eta(t) = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})\left(1 + \cos\left(\frac{t\pi}{T}\right)\right) \qquad (2)$$

To address the class imbalance in the dataset, CrossEntropyLoss is used with class weights. These weights emphasize the underrepresented classes, ensuring the model's predictions are not biased towards more frequent categories.

$$L(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c}) \qquad (3)$$

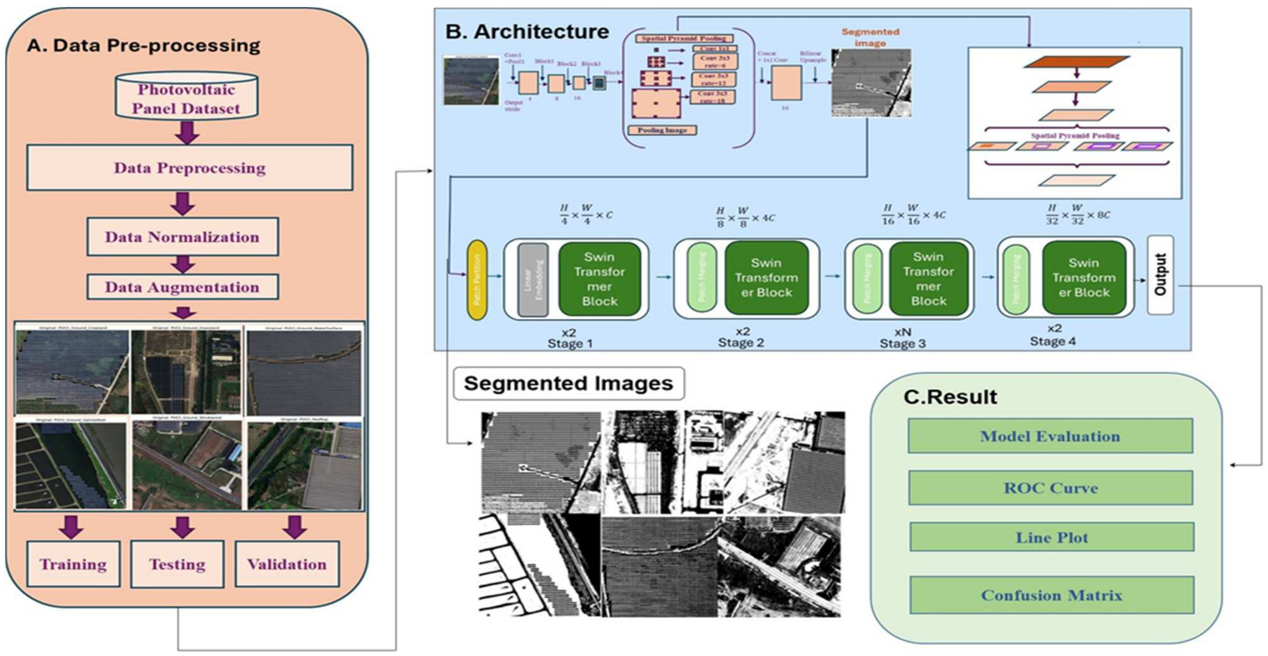


Fig. 1: Overview of Proposed Methodology

Dropout is integrated during training as a regularization technique, which randomly deactivates neurons in the network, forcing the model to learn more robust and generalized features. Together, these techniques ensure the model achieves high performance on both seen and unseen data.

III. Result And Discussion

This research paper presents the results of segmentation and classification for the developed DEEP-SN model of our solar panel detection. The performance of the hybrid model and some other traditional models have been evaluated using the metrics detailed in this section.

A. *Hyperparameters and Training Setup*

All models were trained using the Adam optimizer for 30 epochs, with early stopping based on validation loss and validation accuracy. Key hyperparameters are summarized in Table I.

TABLE I: Summary of training hyperparameters used for Models.

Hyperparameter	Value
Initial Learning Rate	1 × 10−3
Optimizer	Adam
Weight Decay	1 ×10−4
Batch Size	16
Epochs	30
Dropout Rate	0.3
LR Scheduler	OneCycleLR
Class Weighting	Binary cross-entropy

B. *Segmentation Performance Evaluation*

Quantitative evaluation of segmentation performance was carried out using three standard metrics on the test set: mean Intersection over Union (mIoU), Pixel Accuracy (PA), and Boundary F1 Score (BF1). These metrics reflect the overall segmentation quality and boundary precision on solar panel imagery. As shown in Table II, DeepLabV3 demonstrates significantly higher mIoU and pixel accuracy compared to UNet, indicating better overall region segmentation. Despite a marginally lower BF1 score, DeepLabV3 is favored for its ability to capture fine-grained contextual features and leverage a larger receptive field, making it more suitable for complex and high-resolution image segmentation tasks.

In evaluating a solar panel detection model, it is critical to choose the right performance metrics for an accurate analysis. In this context, precision, recall, and the F1 score serve as key metrics.

TABLE II: Segmentation metrics comparison between UNet and DeepLabV3.

Model	MeanIoU	Pixel Accuracy	Boundary F1
UNet	0.3602	0.6665	0.1013
DeepLabV3	0.5795	0.7248	0.1006

The evaluation of the models used in this research reveals significant differences in performance across the various architectures. Among all the models tested, the DEEP-SN model delivered the best results, achieving a precision of 90%, recall of 90%, and an F1- score of 90%. These metrics indicate that the model is particularly robust in accurately detecting the conditions within the dataset. Its performance surpassed other models by a notable margin, which suggests that its architecture likely enhanced by its ability to focus on specific image regions through segmentation was highly effective for this task.

Table III shows that the DEEP SN model achieved a validation accuracy of 91%, outshining several other models, including DenseNet121, InceptionV3, and VGG19. It demonstrated high precision, recall, and F1-score across most classes, with closer scores of 1.00 for conditions such as rooftop and Ground SalineAlkali. However, for certain conditions like Ground Grassland and Ground Shrubwood, both precision and recall were slightly lower, though the overall performance remained strong. Compared to other models, the DEEP SN model outperformed them, with DenseNet121 achieving 64% accuracy and Xception performing poorly with 61% precision.

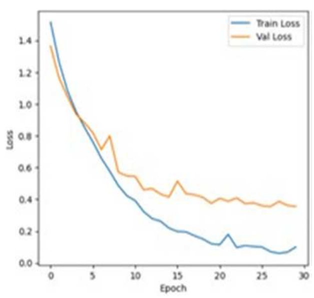
TABLE III. Performance Metrics According to Model

Model	Precision	Recall	F1-score	Support
Proposed DEEP-SN	90%	90%	90%	924
DenseNet121	64%	61%	65%	924
InceptionV3	66%	63%	63%	924
MobileNetV3	61%	58%	58%	924
NASNetLarge	63%	63%	63%	924
ResNet50	71%	67%	70%	462
InceptionResNet101v2	62%	62%	54%	462
VGG16	83%	84%	83%	924
VIT	74%	69%	73%	924
Xception	61%	59%	61%	924
YOLOv11	86%	76%	49%	293

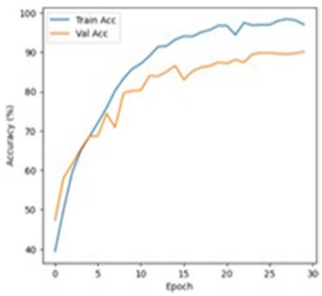
Figure 2 (a) and (b) illustrate the training and validation loss and accuracy trends of the proposed DEEP-SN model. These plots present the steady decline in loss and the gradual increase in accuracy over multiple epochs.

In both subfigures, the x-axis represents the number of training epochs, while the y-axis represents the loss and accuracy values, respectively. The training loss decreased rapidly before stabilizing, while the validation loss exhibited a similar trend but remained slightly higher. This indicates that the model effectively avoided overfitting or underfitting, achieving a strong generalization to unseen data. The training accuracy consistently improved, reaching up to 90%, while the validation accuracy followed a parallel trajectory, demonstrating the model's ability to generalize well to new inputs.

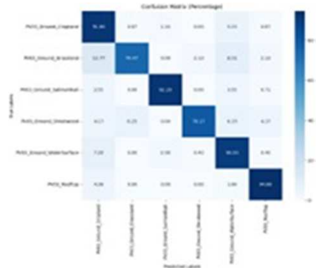
The confusion matrix in Figure 2 (c) further validates the model's performance. It shows that conditions like rooftop and Ground SalineAlkali were classified with perfect accuracy. However, certain classes, such as Ground Grassland, showed minor misclassifications, as evidenced by the off-diagonal values. The ROC curve in Figure 2 (d) underscores the model's discriminative ability, with AUC values nearing 1.0 for all classes. This strong performance highlights the effectiveness of the DEEP-SN model in distinguishing between various conditions in the dataset, making it a reliable tool for this classification task.

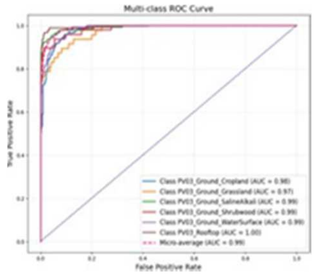


a. Train and validation loss



b. Train and validation accuracy


c. Confusion matrix



d. ROC curve

Fig. 2: Performance evaluation of the proposed DEEP- SN model

IV. CONCLUSION

The DEEP-SN model demonstrated strong performance in classifying solar panel images, achieving a 90% accuracy and outperforming baseline models such as DenseNet121 and Xception. Its effectiveness in detecting specific panel types—particularly rooftop and Ground Saline-Alkali—highlights its robustness and generalization capability, as reflected by the minimal gap between training and validation losses. While certain categories may require further optimization, the model presents a reliable solution for complex classification tasks. Nonetheless, the study is limited by the dataset's size and variability, which may constrain its generalizability. Future work will focus on expanding the dataset to include a wider range of panel types and environmental contexts. Enhancing the architecture with techniques such as attention mechanisms may further improve accuracy. Real-world deployment and validation will be essential to assess the model's practical utility and facilitate broader implementation.

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